

9.6 Appendices

Appendix 9.A: The ESPRIT Algorithm

An important example for high-resolution algorithms is the *ESPRIT* (*Estimation of Signal Parameters by Rotational Invariance Techniques*) algorithm. The basic idea is as simple as it is ingenious. Consider an antenna array with N_r elements. Now create a subarray with the elements $1, 2, \dots, N_r - 1$, and a subarray with elements $2, 3, \dots, N_r$. It is clear from Sec. 8.4.1 that the steering matrices of the subarrays are linked by the following matrix equation

$$\mathbf{A}_2 = \mathbf{A}_1 \Phi \quad (9.52)$$

where Φ is a diagonal matrix whose main-diagonal entries are $\exp(-jk_0 d_a \cos(\phi_i))$. Define now a selection matrix $\mathbf{J}_k$ so that $\mathbf{J}_k \mathbf{A} = \mathbf{A}_k$; this allows to write Eq. (9.52) as

$$\mathbf{J}_1 \mathbf{A} \Phi = \mathbf{J}_2 \mathbf{A} \quad (9.53)$$

In the next step we compute the signal space, i.e., the vector space spanned by the receive vectors $\mathbf{r}_r$ spaces, see [Strang 2016]). A basis for those vector spaces can be obtained by an eigenvalue decomposition of the covariance matrix

$$\mathbf{R}_{rr} \approx \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\dagger \quad (9.54)$$

where the columns of the matrix $\mathbf{U}$ are the basis of the signal space, and $\mathbf{\Lambda}$ is a diagonal matrix. The equation becomes exact when the noise is negligible.

If the signal is noisy, the basis for the signal subspace is defined by the columns of $\mathbf{U}$ that correspond to the signal subspace $\mathbf{U}_s$. In practice, these correspond to the K dominant eigenvalues of $\mathbf{R}_{rr}$ (finding the number of incident waves N , also called “source order”, is nontrivial; for a discussion see, e.g., [Haardt and Nosssek 1994]).

Assume now that N un-correlated plane waves impinge on a ULA of N_r sensor elements and $N \leq N_r$. The eigenvalue decomposition of the covariance matrix is given as

$$\begin{aligned} \mathbf{R}_{rr} &= \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\dagger \\ &= \mathbf{U}_s \mathbf{\Lambda}_s \mathbf{U}_s^\dagger + \mathbf{U}_n \mathbf{\Lambda}_n \mathbf{U}_n^\dagger \\ &= \mathbf{U}_s \mathbf{\Lambda}_s \mathbf{U}_s^\dagger + \sigma_n^2 \mathbf{U}_n \mathbf{U}_n^\dagger \end{aligned}$$

Therefore the basis for the signal subspace is defined by either the columns of $\mathbf{U}_s$ or, equivalently, only those columns of $\mathbf{U}$ which correspond to the N dominant eigenvalues of $\mathbf{R}_{rr}$.

It is also obvious that the matrix $\mathbf{A}$ is in the signal space spanned by $\mathbf{U}_s$. Thus there exists a matrix $\mathbf{T}$ so that $\mathbf{A}$ can be represented as $\mathbf{A} = \mathbf{U}_s \mathbf{T}$. Thus, Eq. (9.53) can be written as

$$\mathbf{J}_1 \mathbf{U}_s \Psi = \mathbf{J}_2 \mathbf{U}_s \quad (9.55)$$

where

$$\Psi = \mathbf{T} \Phi \mathbf{T}^{-1} \quad (9.56)$$

The matrix Ψ is created from the matrix Φ by a transformation that preserves the eigenvalues. In other words, if we know the eigenvalues of Ψ , we also know the eigenvalues of Φ . The eigenvalues of Φ give the angles of incidence via the relationship $\exp(-jk_0 d_a \cos(\phi_i))$. This completes the search for the DOAs.

Thus, the ESPRIT algorithm uses the following steps:

- Determine the covariance matrix of the received signal, $\mathbf{R}_{rr}$.
- Perform an eigenvalue decomposition, and separate signal subspace from noise subspace, in order to determine the eigenbase $\mathbf{U}_s$.
- Determine the matrix $\mathbf{\Psi}$ by solving the (usually overdetermined) system of equations 9.55.
- Find an eigendecomposition of the matrix $\mathbf{\Psi}$
- Determine the directions $\phi_i = \arccos[\arg(eigval(\mathbf{\Psi})) / (-k_0 d_a)]$.

It has to be stressed that ESPRIT relies on the shift invariance of the antenna structure, and thus is mainly suitable for measurements with regularly-structured where all elements have the same patterns.

This description covers only the basic principles. Details of the implementation, as well as generalizations to the multi-dimensional case, estimation of the number of incident signals, etc., can be found in [Haardt and Nossek 1995].

Appendix 9.B: Guidelines for Evaluation of Channel Measurements

Performance of channel sounding campaigns is a large effort, both in terms of cost and work effort. Thus, careful preparation and evaluation is essential to extract the maximum insight from those results. This appendix describes guidelines that in the experience of the author are useful. The emphasis here is on "conventional" evaluations based on Fourier analysis; for evaluations using high-resolution parameter estimation, see Sec. 9.5.3-9.5.8. Please also note the disclaimers stated on the first pages of the book.

Measurement preparation

In the planning of the measurements, it is firstly essential to clarify the goals. For concreteness consider the scenario of measuring a cellular system, with the sounder transmitter (TX) located at the base station (BS) location, and receiver (RX) at the user equipment (UE) location. Some of the key questions to clarify might include

- What is the frequency range and bandwidth of interest? If we are, e.g., measuring LTE where 20 MHz is the maximum bandwidth: do we just need the channel characteristics within this band? Or is the goal the identification of correlation between adjacent bands, or even between uplink and downlink bands, in which case measurements over hundreds of MHz might be needed?
- What is the distance range between BS and UE of interest?
- Are directional characteristics required?
- Is a measurement at a few select locations ("drops") sufficient, or is the evolution of the channel as the UE moves on a trajectory desired?
- Is the measurement to be performed with or without moving IOs - e.g., is one of the goals the identification of time variance that arises from passing cars?

These goals may determine what measurement equipment can be used. For example, if measurement of time variations from a passing car is desired, then a channel sounder based on a VNA will usually not be an option. For long-distance measurements, powerful transmit amplifiers, and/or sounders with large time-bandwidth products, may need to be used. Investigation of directional characteristics requires arrays (real, switched, or virtual), and so on. Further constraints on the measurement setup might arise from regulatory constraints on frequency (channel sounding measurements cannot be done in actively used cellular bands unless the operator owning those bands agrees), power and power spectral density, access to buildings (especially rooftops), and so on. When people are in the vicinity of the TX, attention must be paid to the SAR limits.

Also required is the planning of the exact locations at which the UE shall be placed for the individual measurements.

- Should multiple measurements be performed within a region of stationarity, so that, e.g., small-scale fading can be investigated? It then needs to be decided whether to place the measurement locations on a line (which is typically easier to realize when, e.g., the UE is on a mobile cart that can be moved along a line), or on a rectangular grid (which provides more measurement points within a stationarity region).
- What is the minimum spacing of the points to obtain good decorrelation? The $\lambda/2$ rule only applies for omnidirectional antennas with uniform azimuthal power spectrum of the signals. Does a movement along a line provide a sufficient number of decorrelated points within a stationarity region? The required number of spatial points with essentially uncorrelated fading is also related to the bandwidth of the measurements, since multiple independent samples can

be obtained per measurement location if the measurement bandwidth is significantly larger than the coherence bandwidth.

- When measurements are done in multiple stationarity regions (as is usually the case), what should the placement of those regions be? This question will also be discussed below when describing pathloss fitting methods.

While the above discussion assumed that the goals of the measurement campaign will determine the characteristics of the sounder and the choice of measurement locations, in many situations it might be the other way around: a sounder is available, and will determine what can be meaningfully measured. This is particularly true for state-of-the art equipment that might take many months to construct and require expensive components.

Calibration

The next step in the preparation of the measurement campaign is the calibration of the channel sounder. The goal is typically the determination of the transfer function of the sounder; measured channel transfer functions are divided by this calibration transfer function to eliminate the influence of the equipment from the measurement.

Calibration generally falls into two categories: calibration of the sounder hardware, and calibration of the antennas. For the sounder calibration, the TX and the RX connectors are connected with a cable (and - if necessary - a calibrated attenuator, to avoid damage to the receive equipment due to power overload). The impulse response (transfer function) of the equipment is then directly measured. Note that due to temperature drift, this impulse response might change over time; it is thus typically advisable to let the equipment "warm up" (reach a thermal steady state) before performing the calibration.

The calibration of the antennas is ideally performed in an anechoic chamber. Two types of chambers exist, far-field and near-field. The former has to be large enough so that the antenna under test is in the farfield (compare Chapter 8) of a reference antenna. In the latter case a mathematical transformation can be used to get from the measured near-field values to the far-field patterns. If a chamber is not available, measurements in an actual "greenfield", possibly augmented by delay gating, can be a reasonable substitute. In either case, the goal is the measurement of the (frequency-dependent) antenna gain as a function of angle. This is important not only for directionally resolved measurements, but also for omnidirectional measurements, since in this case a "mean effective gain" can be computed from the (approximate or assumed) knowledge of the angular power spectrum and the antenna gain.

Antenna calibration is a science of its own. The "best practices" document of IEEE on how to measure, and what characteristics the anechoic chambers or outdoor ranges have to fulfill, is an excellent starting point for such measurements [IEEE 2021]. Measurements can be performed either with the help of a reference antenna with known gain at one link end, via the two-antenna method (which assumes that two antennas with identical characteristics are available), and the three-antenna method, which is more general in that it does not assume existence of two identical antennas, but also more cumbersome. Note that the antenna characteristics provided on data sheets are rarely sufficient as "calibration". Also, it is interesting that even nominally omnidirectional antennas can have gain variations of several dB over the 2π radians of azimuth.

The reason why sounder calibration and antenna calibration are performed separately is that antenna characteristics are difficult to measure, but do not vary over time, while the sounder characteristics are easy to measure but might vary over days or weeks. Thus, the sounder characteristics can be easily re-measured at regular intervals, and combined with the antenna characteristics.

During the measurements, it is recommended to have (i) cameras running to identify possible effects that might impact the measurements, like passing cars (of course, such recording is subject to permits at the locations and privacy regulations that might be applicable), (ii) record possible interferers with a spectrum analyzer, and (iii) carefully record the locations of the mobile part of

the sounder as a function of precise time, so that each measurement can be exactly linked to the location at which it was taken.

Pathloss evaluations

Most measurement campaigns will provide measurements of the path gain and shadowing. The evaluation of those results will depend on the type of model whose parameters are to be fitted (e.g., street-by-street vs. cell-wide; or pathloss as function of Euclidean or Manhattan distance). For concreteness we discuss in the following the most popular model, namely an $\alpha - \beta$ fit of the pathloss as a function of the Euclidean distance, applied to the ensemble of all measurement points.

Averaging of the small-scale fading The first step in the evaluation procedure is the averaging-out of the small-scale fading. As discussed above, this can be done through averaging over space (within a stationarity region) and/or frequency (within a stationarity bandwidth). Pitfalls that need to be watched out for are

- the averaging needs to be performed over the received power (not over the complex transfer functions, which often are zero-mean).
- If the size of the ensemble over which the averaging is performed is insufficient, then the variance of the measured points around the straight-line fit (see below) is not purely due to shadowing. Rather, the fading distribution for each stationarity region is typically a variation of the Suzuki distribution, namely a lognormal-gamma distribution, with the lognormal distribution describing the shadowing, while the gamma distribution (whose order is the diversity order available in the ensemble) describes the small-scale fading. Measurements along a trajectory typically have a smaller number of spatial points within the stationarity region, so that the contribution of the small-scale fading is larger, and at the same time the shadowing along such a trajectory is smaller. Still, it is common in the literature to neglect - for all types of measurements - the residual impact of the small-scale fading and ascribe all variations around the straight-line fit to shadowing.
- The measured power is not the signal power (corresponding to the path loss), but rather the sum of signal and noise power. Thus, for points that are in the small-scale fading dip, the "raw" measured power is dominated by the noise power, leading to overestimation of the average channel gain. On the other hand, if noise thresholding (see below) is used, the power is set to zero, which can lead to an underestimation. More on this problem will be discussed below under the catchword of "censoring".

Drops vs trajectories; weighting of points Measurements underlying pathloss evaluations can be done on trajectories, or at random drops within the area of interest. In the latter case, spatial correlations can be analyzed relatively easily, while for random drops this is only possible under specific circumstances (dense spacing of the stationarity regions). Generally, the question of where to place measurement points is a very important one, as discussed next.

If the measurements should investigate, e.g., the dependence of the channel characteristics on the BS-UE separation, then the stationarity regions should be spaced somewhat uniformly along the distance axis. However, even this seemingly simple prescription is loaded with difficulties. Take, e.g., the measurement of power as a function of distance, where we know that the pathloss in dB is a linear function of $\log(d)$. Thus, should the stationarity regions be spaced uniformly along d , or uniformly with respect to $\log(d)$? Or should the points be spaced with a density d^2 , corresponding to a uniform area density arising from a random dropping of UEs in an area? There is no "right" answer to this question, but the pros and cons of the various approaches should be weighted for the particular goals of the measurement campaign, see [Karttunen et al. 2016].

Location of the measurement points will also be governed by the practicalities of where it is even possible to place the measurement setup - for example, it will usually not be possible to put a measurement device in the middle of a busy intersection. There might generally be areas (at particular distances) where measurements can be done more easily, and thus stationarity regions are more concentrated. This can bias the pathloss fitting results. The "local" slope of the pathloss fit can be different at different distances d , either because of physical effects, or just due to statistical variations. Consequently, the distance at which measurement points are concentrated can have a disproportionate impact on the pathloss fit.

The weights w for the pathloss fit (9.58)-(9.59) can, e.g., be chosen to be inversely proportional to the number of data points in a bin:

$$w = \frac{1}{N_b} \frac{N_s}{N}, \quad (9.57)$$

where N_b is the number of data points in the bin and N_s is the total number of points. Adjustments to avoid "overweighting" of distance bins with small number of measurement points can be made, e.g., by choosing a maximum allowed weight level.

Censored data In experiments, the observation of the received signal power at the RX is limited by the system noise, and signals with power below the noise floor can not be measured. This effect can be especially severe at higher frequencies, since the path loss and large scale fading is in general greater compared to lower frequencies. In path loss measurements, samples that are dominated by noise are sometimes referred to as *outage samples*.

It is tempting for an experimenter to simply discard measurements in which the results are below the noise floor (or the measurement system cannot even synchronize, if over-the-air or trigger-based synchronization is used). However, this would throw away valuable information, namely that the pathloss is below the sensitivity threshold of the sounder, and thus the measurements are censored. Also attention has to be paid to the selection of measurement locations. When they are chosen on a regular grid (including regularly spaced on a trajectory), or at completely random locations, then the extracted pathloss truly characterizes the environments. Selecting only those locations for measurement where one can expect reasonable receive power can lead to a selection bias in the pathloss results.

The outage samples should be taken into account in one of two ways:

- An explicit modeling of how many locations are in outage. Obviously this percentage will depend on the distance.
- Adjust the computation of the pathloss coefficient and shadowing variance to take the outage into account. This gives the correct result if the shape of the underlying fading distribution is known and taken into account. The usual assumption here is that shadowing is lognormal (even the combination of small-scale and large-scale shadowing can be approximated as lognormal, see Sec. 5.8).

Note that the estimation of parameters of censored data is not only useful for pathloss estimation, but also cross-polarization discrimination and other channel parameters. In any case, parameter estimates can be improved if the outage samples are modeled as censored samples, and then taken into account in the estimation step.

Parameter estimation The simplest way of estimating the parameters is a linear regression, i.e., determination of a linear fit to the pathloss data (on a dB scale) as a function of $\log_{10}(d/d_0)$, and a subsequent estimation of the shadowing variance as the variance of the measured pathloss values around that linear fit. More generally the formulation can be based on the log-likelihood function. In order to correct for the unintentional bias that is caused by the distribution of data as

a function of the distance, distance dependent weights are introduced in the typical log-likelihood functions:

$$LLF = \sum_{i=1}^{N_s} w(d_i) \left(\log \frac{1}{\sigma^2} + \log \left(\phi \left(\frac{PL_d(d_i) - PL_m(d_i)}{\sigma} \right) \right) \right), \quad (9.58)$$

where N_s is the number of uncensored data samples, σ is the standard deviation (std) of the distribution, w is the weight, PL_m is the path loss model and PL_d is the path loss data at distance d_i given in dB. Here, $\phi(\cdot)$ is the standard normal probability density function (PDF).

If censored data samples are considered, the log-likelihood for these samples have to be considered as well and it is given as

$$LLF^* = \sum_{i=1}^{N_s^*} w(d_i) \left(\log \left(1 - \Phi \left(\frac{PL_d^* - PL_m(d_i)}{\sigma} \right) \right) \right), \quad (9.59)$$

where $*$ refers to censored data, i.e., N_s^* is the number of censored data points, the path loss level for the censoring is PL_d^* , and $\Phi(\cdot)$ is the cumulative distribution function (CDF) of the standard normal distribution. The path loss parameters are then estimated as the minimum of the negative of the log-likelihood function as

$$\arg \min_{\alpha, \beta, \sigma} \{-(LLF + LLF^*)\}. \quad (9.60)$$

Generalizations to dual-slope models and distance-dependent shadowing variance is straightforward [Karttunen et al. 2016]. Also note that it is possible to introduce constraints on the optimization parameters to accommodate possible side information.

It is important to realize that the estimate of a pathloss coefficient is just that - an estimate. Interpreting the shadowing as Gaussian-distributed errors of observations that "should" be on a line, it is possible to ascribe a confidence interval to the linear fit (we use here again the simple linear fit procedure for obtaining the $\alpha - \beta$ model parameters); typically, the 95% confidence interval is provided. Similarly, the parameters we will describe below should be characterized by confidence intervals. Most standard mathematical and statistical packages, including MATLAB, provide confidence intervals estimation, so we refrain here from a detailed discussion. However, it is worth remembering that qualitative statements about channel characteristics (e.g., whether angular spread increases or decreases with distance) needs to be seen in this context - if the confidence interval of the slope includes both positive and negative values, then no confident assessment can be made, even if the mean (i.e., the "standard" regression slope) is, e.g., positive.

Fading distribution evaluation

The evaluation of the small-scale fading statistics consists of two fundamental, inter-related parts: (i) defining which distribution to fit the data to, and (ii) determining the optimum parameters of the distribution that fit the observed values. Generally, the "correct" distribution is not known a priori, and thus has to be determined as part of a fitting procedure. Thus, it is common to define a set of possible distributions, determine the optimum parameters for each of them, and then see which of those optimized distributions fits the measurement data best.

Estimation of the distribution parameters from the measurements depends on the particular distribution at hand. For example, for the Rayleigh distribution, the only free parameter is the mean power (or its equivalent), which can be estimated in a straightforward way. The situation becomes more complicated for two-parameter distributions such as the Rice distribution.

Parameter estimation generally falls into two categories: (i) moment-based estimation, and (ii) maximum-likelihood estimation. Moment-method based estimation generally is simpler, while maximum-likelihood provides the more accurate results. Examples are the Rice-factor estimators based on method of moments [Greenstein et al. 1999] and maximum-likelihood [Chen and Beaulieu 2005]. Further estimators exist for Nakagami, Weibull, $\kappa - \nu$ and many other distributions.

To choose between different distributions (with optimized parameters), two approaches are commonly used: (i) the Kolmogorov-Smirnov (K-S) test, and (ii) the Akaike Information Criterion (AIC).

The K-S test checks the agreement between the empirical cdf and the theoretical distribution. Specifically, the standard K-S test analyzes the parameter

$$D = \sup_z |F_z(z) - F_0(z)|. \quad (9.61)$$

The null hypothesis, i.e., the hypothesis that the distribution follows the theoretical distribution, is rejected if D is larger than a critical value obtained from a table; such a table provides, for a given theoretical distribution and a given significance level, the critical value. Furthermore the critical values are different depending on whether the parameters of the theoretical distribution are fixed a priori or whether they are estimated from the empirical data [Babu and Rao 2004]. Again, many software packages including MATLAB provide various forms of K-S tests. Note the important formulation that "success" just means that we do not reject the null hypothesis at the given significance level; we do not "accept" the null hypothesis. In any case, the K-S test is often used for model selection in that the model with the better pass rate or lower critical value.

The AIC is an estimator for the relative quality of a statistical model, and thus can be used for model selection. It is based on the fact that for large sample sizes (n) the double expectation $E_y \left\{ E_x [\log(g(x|\hat{\theta}(y)))] \right\}$, where $\hat{\theta}(y)$ is the set of parameters estimated based on the data, can be estimated by:

$$E_y \left\{ E_x [\log(g(x|\hat{\theta}(y)))] \right\} = \log(\mathcal{L}) - K \quad (9.62)$$

where $\log(\mathcal{L})$ is the maximum log-likelihood function and K is the number of parameters used in the model. The final formula of AIC is

$$AIC = -2 \log(\mathcal{L}) + 2K \quad (9.63)$$

and AIC selects the model that gives the smallest value. When the sample size is small, it is preferable to use a modified criterion

$$AIC_c = AIC + (2K(K+1))/(n-K-1); \quad (9.64)$$

which converges to the original AIC as the sample size n increases. Since an individual AIC value cannot be interpreted, it is often preferable to analyze $\Delta_i = AIC_i - AIC_{min}$ where AIC_{min} is the minimum value of AIC from all the possible candidates. The greater the AIC difference is the less empirical support the model has. The likelihood of a model is proportional to:

$$\mathcal{L}(g_i|x) \propto \exp\left(\frac{-\Delta_i}{2}\right) \quad (9.65)$$

Finally, for a better interpretation of the likelihood of a model, the Akaike weights are the normalized values of the likelihood $\mathcal{L}(g_i|x)$:

$$w_i = \frac{\exp\left(\frac{-\Delta_i}{2}\right)}{\sum_{p=1}^P \exp\left(\frac{-\Delta_p}{2}\right)} \quad (9.66)$$

Using the same approach, we can find the Akaike weights for AIC_c . As can be seen, AIC offers a simple way to select a model which is closest to the true model in terms of the KL information loss.

Power delay profiles and delay spread

We now turn to the characterization of the delay dispersion. We will assume that measurements of the transfer function are available, i.e., $H(f_i)$. The goal is the determination of power delay profile, rms delay spread, and/or window parameters (compare Chapter 6.4). We stress again that these deliberations apply to Fourier analysis of the measurements. while for HRPE evaluations, equations for delay spread and other parameters can be directly applied to the extracted MPCs.

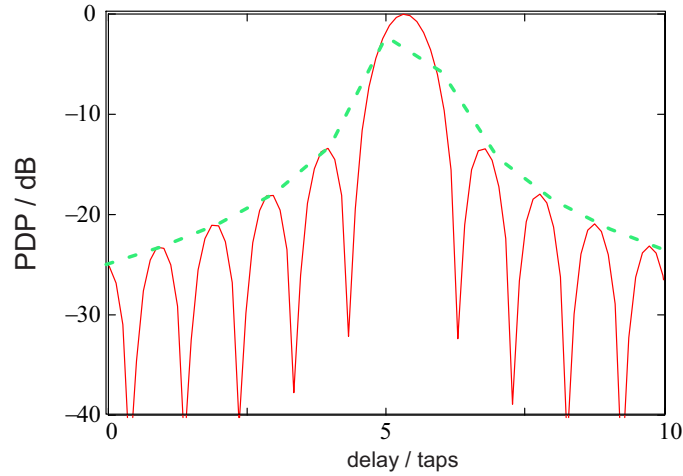


Figure 9.10: PDP of a single MPC at delay 5.4 when using critical sampling (and linear interpolation between the sampling points in green, and 8-times oversampling in red). Note that the correct continuous PDP can be recovered from the critically sampled values through Nyquist-Shannon interpolation, but direct application of the discrete delay spread equation to the critically sampled values of the PDP gives wrong results.

Frequency-domain filtering and sidelobes The first step is the application of an inverse Fourier transform, which is normally implemented via an IFFT. Two important aspects need to be considered here: (i) filtering and (ii) oversampling. Concerning the filtering, remember (see Chapter 10) that the IFFT of a rectangular function is a sinc function, which shows significant sidelobes: the maximum sidelobe is only 13 dB below the peak, and the envelope of the sidelobes decays as $1/f$. Thus, when inspecting impulse responses or power delay profiles, it is often not clear whether a particular peak represents an actual MPC or is just a sidelobe of a different, stronger, peak. One can thus consider multiplying the measured transfer function with a filter function before the IFFT. A variety of such filters exist, e.g., Hann, Blackman, Tukey, and Kaiser (many of these are available functions in MATLAB). The choice of filters is generally a tradeoff between maximum sidelobe level, sidelobe decay (though, for practical reasons discussed below, sidelobe suppression of more than 30-40 dB is usually irrelevant), and width of the main lobe, which in turn impacts the delay resolution. There is no "best" filter, but rather the choice of the tradeoff will depend on the particularities of the measurement system and the goal of the campaign.

A second, sometimes neglected, aspect is the oversampling. Application of the standard IFFT provides critical (Nyquist) sampling of the impulse response. Now imagine that an MPC falls midway between the critical sampling grid: then the discretized (on the critical grid) impulse response has two peaks of approximately equal magnitude (at the sampled delays to the left and right of the actual delay of the MPCs), resulting in a higher delay spread (and also a discrete impulse response that is more difficult to interpret) than if the MPC falls onto a grid point. For this reason, it is recommended to oversample the impulse response, by a factor, e.g., of 8, and extract the delay spread and other parameters from that. An example is shown in Fig. 9.10.

It must also be remembered that filtering distorts the transfer function. Thus, if the filtered transfer function is used for computation, e.g., of the received power, then one has to (i) ensure proper normalization (e.g., one might aim to maintain the total power integrated over the frequency range), and (ii) be aware that the effective bandwidth of the measurement is reduced. The latter impacts, e.g., the question of how many statistically independent samples (ratio of coherence bandwidth over system bandwidth) can be obtained from a measurement. For a large number of measurements within a stationarity interval the filtering will not impact the statistical properties

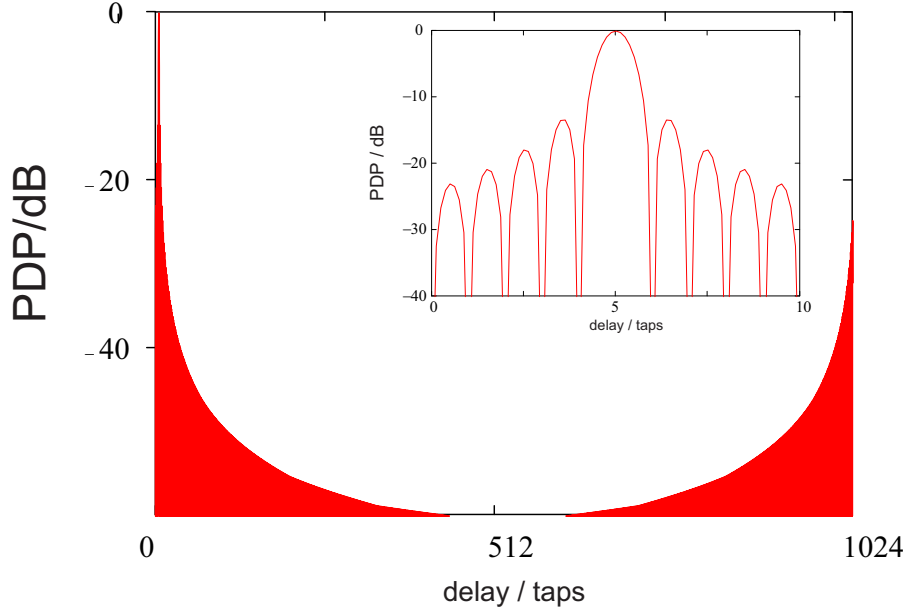


Figure 9.11: PDP of a single MPC at delay 5.4 with maximum excess delay of the sounding system 1024 samples (and the first 10 taps as inset). The wraparound of the LOS component at large delays is clearly visible.

(as long as the system bandwidth is smaller than the stationarity bandwidth), but for a small number of measurements results can differ. Some papers thus use the unfiltered measurement results for power computations, while using filtered results for determination of delay parameters.

Delay gating and noise thresholding The next step in the processing is the computation of the instantaneous power delay profile PDP, i.e., taking the squared magnitude of the impulse response. This PDP can then be averaged over different locations within the stationarity region to provide the average PDP (APDP), compare Sec. 6.5.

As we have discussed in Chapter 6, noise has a major impact on the computation of the rms delay spread and other delay parameters. It is thus essential to perform noise reduction by (i) delay gating and (ii) noise thresholding. Delay gating simply means that the PDP outside an "admissible" delay region is set to 0. The admissible delay region starts slightly before the arrival of the LOS component, and end slightly after the delay of the largest "physically reasonable" MPC. What is "physically reasonable" can often be estimated from inspection of the environment (though care must be taken to account for the possibility of multiple reflections), or from inspection of the raw PDPs. Furthermore, delay gating should eliminate the effect of "wraparound" that is inherent in the use of IFFT: the PDP is cyclically continued; consequently, the precursor sidelobes of, e.g., a sinc function that is centered on a short delay "wrap around" to long delays (compare Fig. 9.11), leading to artificially large delay spreads.

Another important step is the application of noise thresholding. Even with delay gating, any measured PDP will retain a large number of bins in which there is no significant multipath energy, but rather which measures only noise. In order to prevent these noise contributions from distorting the delay spread estimates, a threshold needs to be introduced, such that for all bins in which the power is below that threshold, the power is set to 0. While this prescription seems simple, the choice of the threshold level is a very intricate one, and will be discussed in the following.

First we need to solve the question on how to estimate the average noise power. One possibility lies in turning off the TX from time to time and measuring the RX power during those silent periods

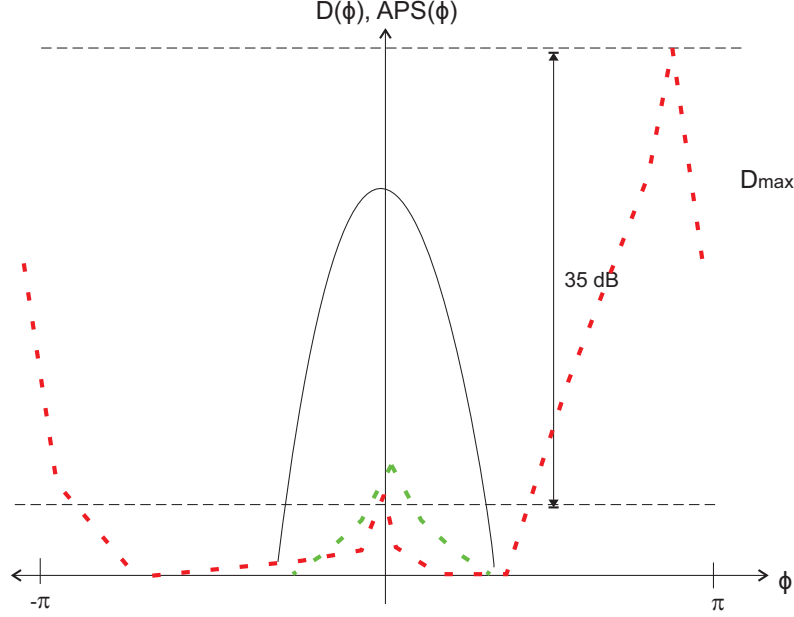


Figure 9.12: Beam pattern pointing to 0 degree: grey solid; APS of (far away, and thus weak) desired user at 0 degree: green dashed; and APS of (nearby, thus strong) interferer: red dashed. It can be seen that a secondary MPC from the interferer near 0 degree (and thus in the beampattern of the desired user) is almost as strong as the desired user, even though this secondary component is 35 dB weaker than the strongest MPC of the same APS.

(we assume here the absence of in-band interference, which is reasonable for channel sounding measurements). If this is undesirable or impossible, the noise level can be estimated directly from the PDP, namely by investigating regions of the PDP that are free from multipath energy; such regions can occur, e.g., for long delay. We will in the following call the estimated noise power per delay bin $\hat{P}_n$. The noise threshold is usually $P_{t,\text{dB}} = \hat{P}_{n,\text{dB}} + \Delta_{n,\text{dB}}$. The choice of $\Delta_{n,\text{dB}}$ determines the probability that a noise peak is not eliminated by the thresholding, and thus impacts the estimated delay parameters (the actual probability for this to occur can be easily computed since the noise power is exponentially distributed). Many papers use $\Delta_{n,\text{dB}} = 6$, but that is only a convention. As a general rule, the noise threshold should increase as the number of delay bins increases (e.g., in wideband measurements), but the thresholding generally constitutes a compromise of eliminating noise and worsening the dynamic range of the measurements. Large-bandwidth measurements at low SNR (e.g., for mm-wave and THz measurements at large distances) are especially difficult because of this.

Another way of setting the threshold is providing a fixed dynamic range, $P_{t,\text{dB}} = \max_k PDP_{\text{dB},k} - \Delta_{\text{DR}}$ using, e.g., $\Delta_{\text{DR}} = 30$ dB. This approach has the advantage of being somewhat simpler (the maximum can be found easily), and the dynamic range can be related to the characteristics of the channel sounder (e.g., quantization noise) and the requirements of system simulations that should be performed with the channel model extracted from the measurements. It can be argued, e.g., that no currently used modulation format needs more than 30 dB SNR for reasonable performance, so that any dynamic range in excess of 30 dB is unnecessary. Note, however, that the latter argument need not be valid in multi-user MIMO systems: due to the near-far effect, a weak component emanating from one UE can lead to strong interference to the signal coming from another UE, see Fig. 9.12.

Finally, in order to ensure that noise does not impact any of the available measurements, Δ_{DR}

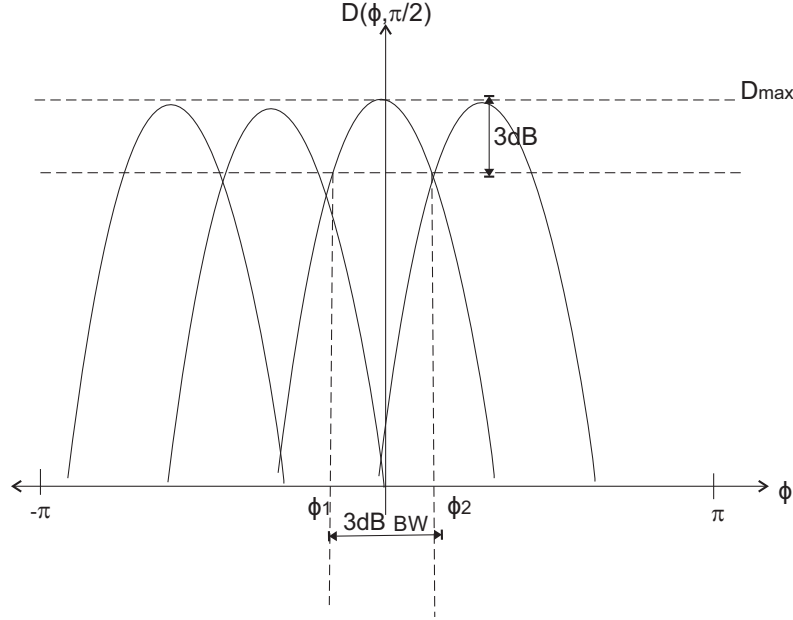


Figure 9.13: Synthesizing an omni-directional pattern from multiple horn patterns.

should be chosen such that $P_{t,\text{dB}}$ is significantly above the noise for *all* measured locations; in other words, the PDP with the worst SNR must determine the choice of Δ_{DR} . A drawback of choosing a fixed dynamic range is that the maximum excess delays evaluated in NLOS situations can become much larger than those in LOS situations, even if the only difference between the two PDPs is the existence/absence of the LOS component.

Angular power spectra and angular spread

Fourier evaluation of the angular characteristics needs to distinguish between the cases that measurements are done (i) by a linear, circular, or rectangular array, or (ii) an array of horn antennas (in either case, the arrays can be real, multiplexed, or virtual). In the former case, the first step is the application of a (spatial) Fourier transformation to get from the spatial (measurement) domain into the angular domain. Just like for the transformation from the frequency to the delay domain, filtering and oversampling should be applied. For this spatial Fourier transformation it is common to not apply any filtering in particular when the antenna aperture is relatively small and the angular resolution thus poor; yet further evaluation has to be conscious of the impact of the sidelobes.

When measurements are done by rotating horn antennas, it is beneficial to have the patterns of the different horns (or horn positions) overlap, such that the -3dB points of the patterns of two adjacent horns coincide (see Fig. 9.13). With such an arrangement, it is easily possible to synthesize (if desired, e.g., for the analysis of omni-directional channel parameters) an omni-directional pattern.

The APS is computed for each delay bin separately, allowing the computation of the ADPS or the DDAPS. It is important that the delay-gated, noise-thresholded versions of the PDPs are used for the computation of the ADPS. Leaving the noise contributions in the ADPS would lead to significant over-estimation of the angular spread (just as they would for the delay spread) and might give a wrong impression about sparsity of the channels in the angular domain.

In either Fourier transform or rotating horn, an MPC might impact multiple angular "bins". The APS thus obtained from a single MPC shows significant angular dispersion. This problem can be combatted by postulating that during each delay bin only a single MPC is active, and selecting

the corresponding MPC as the maximum over the different angles at this delay. This approach yields good results when the measurement bandwidth is very high (usually in mm-wave and THz channels), but it might not be applicable for measurements with the lower delay resolutions that are typical for cm-wave measurements.

From the ADPS, angular spreads can be computed; please see the definitions in Sec. 6.5 for the actual computations. In any case, it is recommended to use the Fleury definition of the angular spread, since it does not suffer from the dependence of the choice of the origin of the coordinate system.

References

Babu and Rao 2004 Babu, G.J. and Rao, C.R., 2004. Goodness-of-fit tests when parameters are estimated. *Sankhya: The Indian Journal of Statistics*, pp.63-74.

Chen and Beaulieu 2005 Chen, Y. and Beaulieu, N.C., 2005. Maximum likelihood estimation of the K factor in Ricean fading channels. *IEEE Communications Letters*, 9(12), pp.1040-1042.

Greenstein et al. 1999 Greenstein, L.J., Michelson, D.G. and Erceg, V., 1999. Moment-method estimation of the Ricean K-factor. *IEEE communications letters*, 3(6), pp.175-176.

IEEE 2021 "IEEE Draft Recommended Practice for Antenna Measurements," in IEEE P149/D11.1 August 2021 , vol., no., pp.1-201, 6 Aug. 2021.

Karttunen et al 2016 Karttunen, A., Gustafson, C., Molisch, A.F., Wang, R., Hur, S., Zhang, J. and Park, J., 2016. Path loss models with distance-dependent weighted fitting and estimation of censored path loss data. *IET Microwaves, Antennas & Propagation*, 10(14), pp.1467-1474.